

Introduction

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CSI 424/524

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UNIVERSITY
AT ALBANY

State University of New York

Welcome to Computer Security!

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- Office Hours: Tuesday/Thursday 4:30pm–5:30pm, Catskill 141 (on Podium), or by appointment

TA Gaurav Pokharel (gpokharel1@albany.edu)

- Office Hours: ?, ? (on Podium), or by appointment

Learning modules

- ① Security basics: threats, principles, policies, and cryptography
- ② Software security: OS and code level
- ③ Web security: threats specific to web applications
- ④ Network security: securing network protocols

Course structure

- Tools
 - Course page: for syllabus and schedule
 - GitHub: for lectures, assignments, and assignment questions
 - Brightspace: for in-class exercises, feedback, and grades
- Textbook & Readings



- Additional readings (links on the schedule)
 - Read assigned chapters/readings before each lecture
- Textbook (end-of-chapter) exercises
 - Posted in `book-exercises` repository
 - Review after each lecture
 - Discuss at the beginning of the following lecture

Course structure (cont.)

- In-class exercises
 - You receive the points for submitting them; graded just as a feedback
- About 10 lab assignments
- Midterm exam (tentative date listed on the schedule)
- Final exam (date listed on the schedule)
- Project
 - Teams of up to four
 - Required for CSI 524
 - Optional for CSI 424 (worth 10% bonus)
 - Instructions will be posted in a few weeks

Lab environment

Setup will be different depending on your machine architecture:

- AMD64 (Windows/Linux/Intel-Mac): Use a prepared virtual machine (VM) that has all the required tools/configurations
 - Using [VirtualBox](#) (free)
- ARM64 (Apple Silicon Mac): Build a virtual machine and deploy the lab tools/configurations
 - Using VMware Fusion (free for personal use)
- Other options (with caveats)
 - Linux OS other than the one provided: you will need to configure it yourself
 - Deploy the VM on a cloud provider: costs money
 - Your primary Unix-like OS: risking corrupting your primary OS

Your first lab is to set up your environment (see [lab01](#) and [guides](#) repositories).

Lab environment (cont.)

- All your lab reports must show your personal prompt as instructed in lab01
 - You use the same VM for the rest of the semester
- What if your VM environment is corrupted?
 - Highly recommended that you keep a backup copy of your setup after lab01, so you can just re-deploy when needed
 - If you deploy a fresh VM, make sure you set it up again according to lab01

Lab assignments

- Released typically a week in advance of the deadline
- Start early. The least you can do is taking a look at it on day 1
- Ask questions
 - Create issues in **your assignment repository** on GitHub
 - Meet us in office hours
- Discuss freely with your classmates
- Submit individually based on your work on your own VM setup
- We collect your repository from GitHub at the deadline
- Grade will be recorded on Brightspace and announced
 - You have 5 business days to create an issue in your assignment repository if there is any issue with your grade
 - No re-grading after this 5-day period

Lecture activities and participation

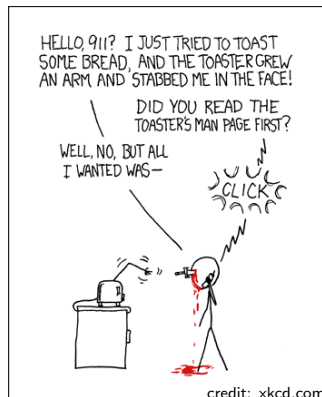
- We will often demonstrate concepts using sample codes
 - I encourage you to have your VM ready during class and try them yourself too
- In-class exercises
 - Discuss them with your classmates sitting next to you
 - Submit on Brightspace
 - You are allowed to submit them in person
 - Submission matters; not the grade!

Policy highlights

- No tolerance for academic dishonesty
- No late submission (generally)
 - Passed 11:59pm on the day of the deadline and you will receive no point
 - All sort of software and hardware problems may occur
 - Plan ahead and do not leave it to the last minute
- Can ask for one-time late submission
 - If granted, you can submit 3 days late
- Lowest lab grade will be dropped
- Up to 10% of in-class exercises will be dropped
- 5-day period to post any issue regarding a grade

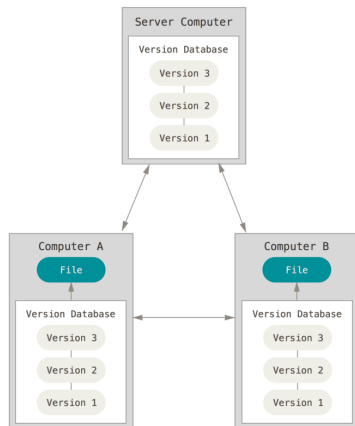
How to succeed in this class

- Read ahead and exercise!
 - Book chapters are easy to read
 - Practice what you read as you read it (or during/after class)
 - Read the freaking manual!
- Start your assignments early
- Get help from us!
 - GitHub, office hours, email, . . .



Git and GitHub

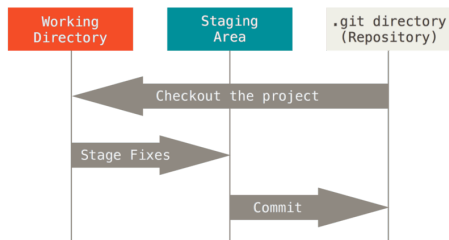
- Git is a distributed version control system
- GitHub is a cloud solution for storing git repositories that acts as one of the distributed nodes



Git states

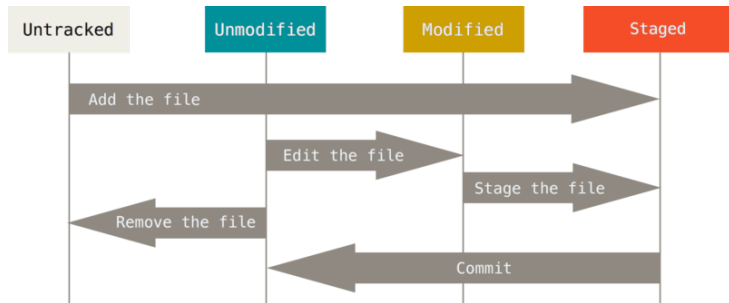
On your local machine files can be:

- In your local repo (committed)
- Checked out and modified, but not yet committed (working copy)
- In-between, in a “staging” area
 - Staged files are ready to be committed
 - A commit saves a snapshot of all staged state



Basic Git life cycle

- **Modify** files in your working tree
- **Stage** files (selectively), adding snapshots of them to your staging area
 - Also, **track** files in your working directory
- **Commit**, which takes staged files and stores that snapshot permanently to your Git directory



Basic Git life cycle: commands

Git sub-commands that you mostly need:

- `init`: create an empty Git repo
- `status`: show working tree status
- `add`: stage a file
- `commit`: record staged changes to repo

Other sub-commands you may use less frequently:

- `log`: show commit logs
- `diff`: show changes between commits, commit & working tree, etc.
- `rm`: remove files from working tree and index
- `checkout`: switch branches or restore working tree files

Git: working with remote repositories

- `clone`: clone a repo (create a local duplicate)
- `remote`: manage set of tracked repos
- `fetch`: download objects from remote repo
- `pull`: fetch and merge with local repo
- `push`: update remote repo

Using Git for lab assignments

- On release date of lab
 - ① There will be a repository available with that lab name (e.g., `lab01`) on GitHub
 - ② Follow the link in repository. It will create a private repo will for you for that specific homework: e.g., `lab01-username`
- For working on your homework, you
 - ① Clone your homework repo (`lab01-username` , not `lab01`)
 - ② Create/edit assignment files
 - ③ Add changes
 - ④ Commit
 - ⑤ Push (and go to [2](#) if needed)
- At the deadline, we
 - ① Clone your repo to our machine
 - ② Grade it